

Ten year responses of Engelmann spruce and a high-elevation Ericaceous shrub community to manual cutting treatments in southern interior British Columbia

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Abstract

Vegetation management treatments are often applied to high-elevation, Ericaceous shrub-dominated sites in southern interior British Columbia to facilitate reforestation. We studied the effects of manual cutting on an established vegetation community and newly planted to 2-year-old Engelmann spruce (*Picea engelmannii* Parry ex Engelm.) over a 10-year period. Spruce survival was significantly higher in the treatment than the control 3 years after brushing ($p = 0.020$). Although differences in survival continued to be large after 10 years (79% versus 61%), they were no longer significant due to increased between-site variability. Relative height estimates indicated that brushed spruce were taller relative to the vegetation canopy than control spruce throughout the 10 years assessment period, but growth responses were small even where significant. The cutting treatment significantly increased ground-level diameter after 3 and 5 years, but not after 10 years. The loss of statistical significance is attributed to ongoing mortality of small diameter suppressed stems in the control. After 10 years, shrub cover was still lower in the cutting treatment than the control, but height had recovered within 3 years. Manual cutting increased Shannon diversity in the first year post-treatment ($p = 0.039$), after which it returned to control levels. There were no differences in species richness. A competition index based on the first year relationship between spruce and vegetation cover and height (CRH_{TS1}) explained 33% of the variation in spruce diameter 10 years after treatment, and first year total shrub cover (COV_{TS1}) explained 28% of the variation. R^2 values increased slightly over time, suggesting that vegetation competition was still affecting at least some surviving seedlings when they were 10–12 years old. Thresholds for spruce diameter of 139 and 64% were identified for CRH_{TS1} and COV_{TS1} , respectively.

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1. Introduction

Regeneration of high-elevation clearcut sites is challenging (Farnden, 1994) because of wet and cold environmental conditions, a short growing season and the presence of competitive vegetation communities. Over time, the cumulative effects of these factors may reduce seedling vigour to the

point where survival and growth rates are unacceptable in managed plantations (BC Ministry of Forests, 2000). On many high-elevation sites, vegetation management treatments are applied to relieve constraints to seedling establishment, but the extent to which they succeed often depends on the degree to which limitations imposed by light availability and snow-pressure are relieved relative to other uncontrollable factors such as cold, wet soils and low air temperature (Coates et al., 1991; Farnden, 1994).

One of the most common high-elevation communities in southern interior British Columbia is dominated by Ericaceous shrubs, including white-flowered rhododendron (*Rhododendron albiflorum* Hook.), false azalea (*Menziesia ferruginea* Sm.), black huckleberry (*Vaccinium membranaceum* Dougl. ex Hook.) and oval-leaved blueberry (*Vaccinium ovalifolium* Sm.).

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(Kimmins and Comeau, 1990; Boateng and Comeau, 1997). Related Ericaceous shrub communities occur in coniferous forest understories at various locations in the northern hemisphere (e.g., Franklin et al., 1979), and may affect regeneration of coniferous seedlings through competition for resources (Kneeshaw and Bergeron, 1996) or allelopathy (Mallik, 2003). In British Columbia, the Ericaceous shrub community often co-occurs in a mosaic of patches with a herb-dominated community (Caza and Kimmins, 1987) that includes Sitka valerian (*Valeriana sitchensis* Bong.), fireweed (*Epilobium angustifolium* (L.) Scop.), Indian hellebore (*Veratrum viride* Ait.) and a variety of smaller-stature herb species. These communities occur predominantly in the Engelmann Spruce-Subalpine Fir (ESSF) biogeoclimatic zone (Meidinger and Pojar, 1991), which is found at high-elevations in British Columbia to a latitude of approximately 57°N (Farnden, 1994). The shrub-dominated community is associated with drier and more nutrient-poor edaphic conditions than the herb-dominated community, as well as lower levels of disturbance to the forest floor during harvesting and site preparation practices (Haeussler et al., 1990; Kimmins and Comeau, 1990; Boateng and Comeau, 1997).

At full development, the Ericaceous shrub-dominated community creates dense shade that can maintain soil temperatures below critical thresholds for water and nutrient uptake and reduce the availability of light for conifer seedlings (Coates et al., 1991). Spruce is moderately shade tolerant (Wright et al., 2000; Klinka et al., 2000), and although it can survive at low light levels (e.g., 8–10% full sunlight) (Coates et al., 1994; Lieffers and Stadt, 1994; Chen, 1997), extended exposure to such conditions results in reduced radial growth and substantial mortality (Kobe and Coates, 1997; Wright et al., 1998; Biring et al., 2003). Strong relationships have been found between the abundance of overtopping vegetation, light availability and spruce seedling growth (Comeau et al., 1993; Jobidon, 1994). Ericaceous shrubs also commonly cause physical damage to small seedlings (Boateng and Comeau, 1997), presumably through a combination of vegetation- and snow-press.

High-elevation vegetation communities can be managed to some extent by judicious choice of harvesting and site preparation methods, and current practices specify that planting should take place as soon as possible after site preparation. Vegetation management treatments are also often considered necessary to achieve acceptable conifer seedling survival and growth (Boateng and Comeau, 1997). It can be difficult to judge whether these treatments are effective, however, because although Ericaceous shrubs recover slowly from brushing (Coates, 1987), conifer seedling survival and growth responses are variable and slow to manifest due to environmental constraints. Forest companies seek to avoid application of ineffective or unnecessary treatments, not only because of the high cost, but also because of public opposition to treatments that may affect biological diversity (Burton et al., 1992). In 1992, this study was initiated to examine the long-term effects of manual cutting of Ericaceous shrubs on conifer seedling performance. It was part of a large project designed to study ecosystem-specific effects of common operationally-applied brushing methods to 13 vegetation communities in southern

interior BC (Simard, 1993). Here we present 10-year results for manual treatments applied to release Engelmann spruce (*Picea engelmannii* Parry ex Engelm.) seedlings growing amongst an Ericaceous shrub-dominated community. The objectives of the study were to quantify the effects of vegetation management on (i) survival and growth of Engelmann spruce, (ii) the abundance of target vegetation and (iii) species and structural diversity of the plant communities. A fourth objective was to examine relationships between survival or growth of spruce and the abundance of neighbouring shrub or herb vegetation using regression analysis, and a fifth objective was to identify competition thresholds for spruce diameter growth in subalpine forests. We hypothesized that survival and growth of Engelmann spruce would improve in response to sustained reductions in shrub abundance, and that cover of rhizomatous herbs would increase in response to greater light availability (e.g., Boateng and Comeau, 1997).

2. Methods

2.1. Site description

The study included four sites in the Engelmann Spruce-Subalpine Fir zone (Meidinger and Pojar, 1991), located between 49°15'N and 51°12'N, and 118°49'W and 114°43'W, in the Montane Cordillera terrestrial ecozone of British Columbia, Canada (Wiken, 1986). The ESSF zone is characterized by high snow cover and short, cool growing seasons. The snow free period, particularly at higher elevations and on north aspects, can be as little as 110 days, and frosts are possible at almost any time during the growing season (Farnden, 1994). The sites are in the wet, cold (ESSFwc), the very wet, cold (ESSFvc) and the dry, cool (ESSFdk) subzones, which average 1189 mm precipitation and 824 growing-degree days (>5 °C) annually (Lloyd et al., 1990; Braumandl and Curran, 1992). The sites are of south-easterly to northerly aspect and moderate slope (25–44%), and they range in elevation from 1450–1800 m. Soils are Podzols or Brunisols (Soil Classification Working Group, 1998) with silt loam to loam texture. The study sites were clearcut harvested between 1974 and 1986, and planted with container-grown (PSB 415) Engelmann spruce seedlings up to 18 years after the most recent site preparation or harvesting disturbance. Seedlings were newly planted to 2-year-old (age from planting) when this study was initiated. Prior to manual brushing, the Ericaceous shrub-dominated plant community was similar among replicate sites.

2.2. Experimental design and treatments

We used a randomized complete block design (RCBD) with two treatments (manually cut and untreated) replicated on four clearcut sites. Replicating on different sites rather than within a single site allowed us to draw inferences across the broad range of sites on which the manual cutting treatment is commonly applied (McCone and Lively, 1993). This approach imposes some limitations, however, on the power to detect small treatment effects (Simard et al., 2001). On each site, the treatment and control were each randomly assigned to one of

two plots that measured 0.9 ha in size, including a 20 m surrounding buffer zone. Within each treated and control plot, 36 gridpoints were systematically located, with the closest undamaged Engelmann spruce seedling selected as the measurement tree. Circular subplots of 1.78 m radius (10 m²) centred on the measurement spruce were used for all plant community measurements.

Manual cutting was applied in 1992 as part of the operational brushing program for the southern interior of British Columbia. Vegetation surrounding measurement spruce was cut in an average 0.9 m radius, using brush saws or machetes, on dates between mid-July and mid-September.

2.3. Measurements

Measurements were made immediately before brushing (year 0), and 1, 3, 5 and 10 years afterward. Timing of the pre-brushing assessment was dependent on the operational brushing schedule, and post-treatment assessments were conducted between mid-July and early-September (except in year 10 when late assessment precluded collection of target species data on one site and species diversity data on two sites). In each tree-centered subplot, the measurement spruce was evaluated for survival, total height (± 1 cm), root collar diameter (± 0.1 cm), leader length (± 1 cm) and cause of damage. The cause of mortality for dead trees was identified wherever possible. Height:diameter (H:D) ratio was calculated. Seedling vigour (dead, poor, moderate and good) and competitive status (overtopped (shorter than neighbours), threatened (equivalent height to neighbours), and free-of-vegetation (taller than neighbours)) were visually assessed using criteria described by Simard (1993). Cover and modal height of all vegetation, the shrub layer, the herb layer and selected target species were evaluated within the 10 m² subplots. For a species to be selected as a “target” it had to represent a competitive threat to measurement spruce and be present in at least 28 of 36 subplots in each of the treatment and control plots on individual sites. Relative height was calculated as the ratio of subject spruce height to overall modal height of surrounding vegetation.

In years 0, 1, 3 and 5, percent cover of all individual vascular plant species was recorded in four subplots (randomly selected in year 0) within each treatment and control plot. These data were used to calculate species richness (the number of species) and species diversity according to Shannon and Weaver (1949):

$$H1 = - \left(\sum_{i=1} \left(\frac{n_i}{N} \right) \log \left(\frac{n_i}{N} \right) \right) \quad (1)$$

where n is the cover of each species and N is the sum of cover of all species. Eq. (1) was also used to determine structural diversity of the plant community based on eight vegetation groups: shade tolerant conifers, moderately shade tolerant conifers, shade intolerant conifers, broadleaf trees, tall shrubs, short shrubs, tall herbs and short herbs. Conifers were assigned to shade-tolerance classes after Klinka et al. (2000) and shrub and herb species were assigned to height-related classes based on their usual stature in southern interior British Columbia (based on descriptive sources

such as Parish et al. (1996)). Short herbs and short shrubs tended to attain average heights of less than 30 and 120 cm, respectively. Richness (number) of structural groups was also calculated. Treatment effects on the abundance of individual plant species were identified where species cover values increased or decreased by at least 50% in the treatment relative to the control on at least 50% of the replicate sites. Species and structural richness and diversity results cannot be presented for year 10 because late timing of assessments precluded adequate data collection for two of the sites.

2.4. Analysis

All statistical analyses were performed using SAS for Windows, version 8.2 (SAS Institute Inc., 2001). Conifer, target vegetation and plant diversity continuous variables were compared between treatments for each year using a mixed-model analysis of variance (ANOVA) with four-fold replication in a randomized complete block design ($n = 4$). Block means were based on 36 subplots per treatment unit, but subplots did not contribute to the experimental error used to test treatment effects (Kirk, 1982; McCone and Lively, 1993). Transformation was not necessary for any of the variables based on examination of the distribution of residuals, tests for normality and tests for equality of variance. Repeated measures analysis was not performed because of the unequal time span between measurement years.

Stepwise logistic regression analysis was used to screen ten independent vegetation abundance variables for their correlation with 10 year spruce survival. The independent variables included cover and height of surrounding herbs and shrubs (based on measurements within 1.78 m subplots) in the first year post-treatment. Non-linear and linear regression analyses were used to examine relationships between spruce diameter in years 3, 5 or 10 after brushing and the abundance of surrounding vegetation in year 1 after brushing (the time when treatment effect was strongest). Measures 1 year after brushing included total herb cover (COV_{H1}), total shrub cover (COV_{TS1}), summed cover of Ericaceous shrub species (COV_{ES1}) and the competition index CRH (Brand, 1986; Comeau et al., 1993), calculated using total shrub cover (CRH_{TS1}) or summed Ericaceous shrub cover (CRH_{ES1}). CRH is defined as:

$$CRH = \frac{CR}{H} \quad (2)$$

where C is percent cover of all shrubs or Ericaceous shrubs, R is the modal height of shrubs or Ericaceous shrubs and H is the total height of the measurement spruce. The following non-linear model was used:

$$y = ae^{bx} + \varepsilon \quad (3)$$

where y is a spruce growth variable, x the vegetation abundance variable (or CRH), ε the residual error (assumed to be independent, with the same normal distribution for all plots), a and b are model parameters estimated by the least squares method and e is the base of the natural log (ln).

Competition thresholds were determined for individual sites based on relationships between 3-year spruce stem diameter

and COV_{TS1} or CRH_{TS1} . We selected year 3 diameter response as appropriate for identifying thresholds because (a) spruce had responded significantly to treatment and (b) the shrub community had returned to control levels. The thresholds were identified from a manually fit ceiling function that described the upper boundary of the scattergram and enveloped at least 95% of the observations. This approach was first described by Goldberg (1987), and later applied by others including Wagner et al. (1989) and Burton (1993). The threshold occurred at the vegetation level below which diameter was independent of decreasing cover, and above which diameter declined as cover increased (i.e., the point at which the effects of competition were expressed through reduced diameter growth). An example is provided in Fig. 1.

3. Results

3.1. Spruce survival and vigour

Spruce survival was higher in the treatment than the control 3 years (96% versus 81%; $p = 0.020$) and 5 years (92% versus 71%; $p = 0.079$) after manual cutting (Table 1). After 10 years, when spruce were 10–12 years old, survival was 61% in the control compared with 79% in the treatment, but the difference was not significant due to high between-site variability ($p = 0.261$). Vegetation competition accounted for approximately four-fifths of control mortality and two-thirds of treatment mortality. *Armillaria* root disease and animal damage accounted for less than 1% mortality each, and the remaining seedlings died from unknown causes. Among surviving spruce,

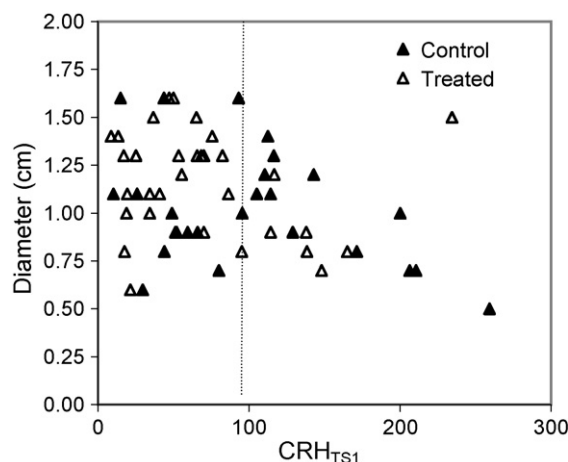


Fig. 1. Relationship between 3 year Engelmann spruce diameter and a competition index based on the first year relationship between spruce and vegetation cover and height (CRH_{TS1}) at a single location (site 106). Data points include both treated and untreated control subplots. The vertical dashed line approximates the competition threshold. Below the threshold, spruce diameter is independent of decreasing shrub cover and height (as defined by CRH_{TS1}), and above it diameter declines in response to increasing CRH_{TS1} .

the proportions that were of good, moderate or poor vigour were approximately the same between the treatment and control in year 10 (Fig. 2).

3.2. Spruce growth

Ten years after manual cutting, spruce basal stem diameter, height, leader length and height:diameter ratio did not differ

Table 1
Summary of one-way ANOVA results^a comparing Engelmann spruce growth variables between manually brushed and control treatments, before (year 0), and 1, 3, 5 and 10 years after treatment

	Survival (%)	Diameter (cm)	Height (cm)	Leader (cm)	Height: diameter	Relative height (%) ^b
Brushing treatment						
Year 0						
Control	100 ± 0	0.47 ± 0.01	26 ± 1	7 ± 0.4	55 ± 1	13 ± 4
Brushed	100 ± 0	0.50 ± 0.01	27 ± 1	8 ± 0.4	54 ± 1	11 ± 4
<i>p</i> -Value	1.000	0.062	0.393	0.841	0.311	0.716
Year 1						
Control	90 ± 2	0.56 ± 0.02	30 ± 1	5 ± 0.3	54 ± 2	22 ± 6
Brushed	98 ± 2	0.58 ± 0.02	30 ± 1	4 ± 0.3	52 ± 2	50 ± 6
<i>p</i> -Value	0.115	0.463	0.760	0.673	0.578	0.038
Year 3						
Control	81 ± 2	0.74 ± 0.01	40 ± 2	5 ± 0.5	55 ± 2	39 ± 7
Brushed	96 ± 2	0.84 ± 0.01	43 ± 2	7 ± 0.5	51 ± 2	58 ± 7
<i>p</i> -Value	0.020	0.008	0.337	0.095	0.221	0.168
Year 5						
Control	71 ± 5	1.00 ± 0.03	53 ± 1	6 ± 0.4	54 ± 1	41 ± 4
Brushed	92 ± 5	1.16 ± 0.03	57 ± 1	7 ± 0.4	50 ± 1	65 ± 4
<i>p</i> -Value	0.079	0.027	0.141	0.095	0.052	0.030
Year 10						
Control	61 ± 9	1.77 ± 0.20	95 ± 3	8 ± 2	59 ± 4	77 ± 3
Brushed	79 ± 9	2.00 ± 0.20	96 ± 3	9 ± 2	49 ± 4	99 ± 3
<i>p</i> -Value	0.261	0.464	0.772	0.956	0.207	0.014

^a Values are presented as "mean ± 1 standard error".

^b Relative height refers to the height of the measurement spruce relative to modal height of neighbouring vegetation.

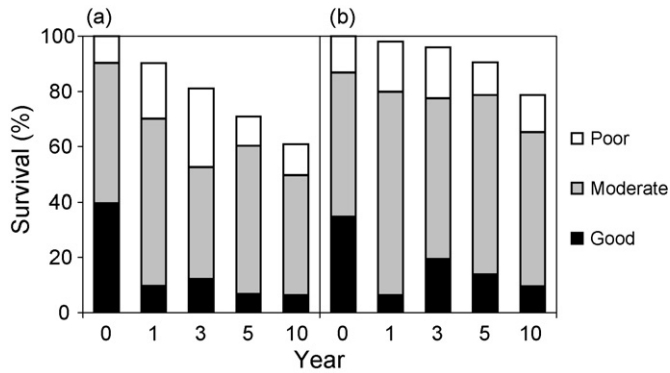


Fig. 2. Mean survival and vigour of spruce in (a) the control and (b) the manual cutting treatment at the time of cutting (year 0) and in years 1, 3, 5 and 10 after cutting.

between the control and treatment (Table 1). Diameter was larger in the treatment 3 years ($p = 0.008$) and 5 years ($p = 0.027$) after cutting, but the differences were small (0.10–0.15 cm), and became non-significant by year 10 ($p = 0.464$).

3.3. Spruce competitive status

In the first year after treatment, brushed spruce were 50% as tall as surrounding vegetation, compared to 22% for control spruce ($p = 0.038$, Table 1). With the exception of year 3, significant difference in relative height continued through year 10, when brushed spruce were 99% the height of vegetation, compared with 77% for control spruce ($p = 0.014$).

Prior to brushing, an average 98% of spruce seedlings were overtopped by vegetation, an average 1% were threatened, and

an average 1% were free-of-vegetation. In the control, this level of overtopping remained constant through year 5, but an average 33% of surviving stems were above surrounding vegetation by year 10. The manual cutting treatment immediately freed the majority of spruce from overtopping vegetation, but more than half were overtopped again by year 3. On average, only 19% of surviving brushed spruce were free-of-vegetation 10 years after manual cutting.

3.4. Vegetation abundance

Manual cutting reduced shrub cover for at least 10 years (Table 2), although the difference was small, and marginally non-significant in years 3 and 5 after treatment. Total vegetation cover followed a similar trend in that differences were significant in year 10, but they were very small (90% in the control versus 87% in the treatment), and were not significant in earlier assessments, possibly because of greater between-site variation. Shrub height was reduced in the first year post-treatment, but recovered to control levels by year 3. Herb cover and height were unaffected by manual cutting and remained relatively constant throughout the 10 year assessment period (Table 2).

3.5. Plant community diversity

Species richness was not affected by manual cutting during the 5-year period after treatment ($p \geq 0.350$). There was an average of 27 species prior to treatment, and 34 species 5 years later. According to the Shannon index, species diversity increased as a result of manual cutting within 1 year (2.50

Table 2

Summary of one-way ANOVA results^a comparing vegetation cover and height variables between manually brushed and control treatments, before (year 0), and 1, 3, 5 and 10 years after treatment

	Cover (%)			Height (cm)		
	Total	Shrubs	Herbs	Total	Shrubs	Herbs
Year 0						
Control	88 ± 4	57 ± 3	49 ± 3	115 ± 7	115 ± 2	53 ± 4
Brushed	88 ± 4	59 ± 3	41 ± 3	111 ± 7	119 ± 2	43 ± 4
<i>p</i> -Value	0.918	0.747	0.138	0.707	0.336	0.185
Year 1						
Control	90 ± 2	60 ± 2	51 ± 3	90 ± 4	97 ± 2	44 ± 8
Brushed	80 ± 2	48 ± 2	51 ± 3	76 ± 4	87 ± 2	29 ± 8
<i>p</i> -Value	0.068	0.020	0.853	0.081	0.038	0.249
Year 3						
Control	84 ± 3	60 ± 3	46 ± 3	113 ± 3	115 ± 5	49 ± 8
Brushed	76 ± 3	48 ± 3	42 ± 3	90 ± 3	95 ± 5	39 ± 8
<i>p</i> -Value	0.137	0.070	0.409	0.014	0.087	0.457
Year 5						
Control	78 ± 3	58 ± 3	44 ± 3	104 ± 6	115 ± 5	35 ± 4
Brushed	72 ± 3	50 ± 3	44 ± 3	87 ± 6	98 ± 5	30 ± 4
<i>p</i> -Value	0.301	0.161	0.970	0.150	0.109	0.471
Year 10						
Control	90 ± 0.5	64 ± 1	40 ± 2	126 ± 17	121 ± 4	31 ± 5
Brushed	87 ± 0.5	55 ± 1	40 ± 2	105 ± 17	121 ± 4	31 ± 5
<i>p</i> -Value	0.031	0.020	0.946	0.465	0.983	0.996

^a Values are presented as “mean ± 1 standard error”. Year 10 means are based on three rather than four sites.

Table 3

Vascular plant species that increased or decreased by at least 50% (by proportion) in the manual cutting treatment relative to the control, from the time of brushing until the fifth year assessment

Increased abundance	Decreased abundance
Shrubs	
<i>Ribes lacustre</i>	<i>Lonicera utahensis</i> S. Wats.
<i>Sambucus racemosa</i> L.	<i>Menziesia ferruginea</i> Sm.
<i>Vaccinium membranaceum</i> Dougl. ex Hook.	<i>Rhododendron albiflorum</i> Hook.
Herbs	
<i>Anaphalis margaritacea</i> (L.) B. and H. f. ex C.B. Clarke	<i>Dryopteris assmilis</i> S. Walker
<i>Veratrum viride</i> Ait.	<i>Epilobium angustifolium</i> L.
	<i>Smilacina racemosa</i> (L.) Desf.
	<i>Streptopus amplexifolius</i> (L.) DC.
	<i>Thalictrum occidentale</i> A. Gray

versus 2.39; $p = 0.039$), but this effect had disappeared by year 3. Trends in the abundance of certain species changed in the treatment relative to the control during the 5-year period after brushing (Table 3). The abundance of three shrub species (two Ericaceous) and five herb species decreased in the treatment relative to the control. During the same period, three shrub species (one Ericaceous) and two herb species increased in the treatment relative to the control.

Manual cutting did not affect the richness of vegetation structural groups, but the diversity of groups was higher in the treatment than the control in year 3 (1.29 versus 1.06; $p = 0.045$). The trend continued through year 5 ($p = 0.055$). The difference is attributed to the presence of trembling aspen (*Populus tremuloides* Michx.) (representing the “broadleaf tree” structural group) in one of the brushed plots.

Table 4

Non-linear^a regression equation parameter estimates for predicting 3, 5 and 10 year spruce diameter from first year neighbourhood vegetation abundance (for all equations $p < 0.001$)

Spruce diameter	Year 1 vegetation abundance variable ^b	<i>n</i>	<i>a</i>	<i>b</i>	Adjusted <i>R</i> ²
Year 3	COV _{H1}	223	0.6612	0.0037	0.044
	COV _{TS1}	223	1.2090	−0.0080	0.230
	COV _{ES1}	217	1.0095	−0.0073	0.163
	CRH _{TS1}	220	1.1401	−0.0021	0.318
	CRH _{ES1}	214	1.0257	−0.0023	0.257
Year 5	COV _{H1}	205	0.8478	0.0047	0.046
	COV _{TS1}	205	0.3059	−0.0134	0.120
	COV _{ES1}	201	0.2433	−0.0145	0.083
	CRH _{TS1}	203	0.3011	−0.0042	0.206
	CRH _{ES1}	199	1.5358	−0.0033	0.291
Year 10	COV _{H1}	181	1.0608	0.0101	0.112
	COV _{TS1}	181	3.7516	−0.0148	0.278
	COV _{ES1}	177	2.7402	−0.0141	0.164
	CRH _{TS1}	179	3.2943	−0.0039	0.331
	CRH _{ES1}	175	2.7938	−0.0045	0.262

^a General form of the non-linear model is: $y = ae^{bx} + \varepsilon$, where y is spruce diameter response, x a neighbourhood vegetation abundance variable and ε is the residual error term.

^b COV_{H1} is herb cover; COV_{TS1} is shrub cover; COV_{ES1} is the summed cover of Ericaceous shrubs; CRH_{TS1} is CRH based on shrub cover; CRH_{ES1} is CRH based on the summed cover of Ericaceous shrubs.

3.6. Regression

Stepwise logistic regression analysis showed a significant relationship between 10 year spruce survival and first year post-treatment vegetation abundance, with herb cover $\times$ height as the independent variable ($p < 0.001$). However, the model accounted for none of the variation in survival between brushing and control treatments (odds ratio = 1.000). In contrast, relationships between spruce diameter and vegetation abundance were significant ($p < 0.001$). Both non-linear (Table 4) and linear (data not shown) models revealed a negative relationship between first year total shrub or summed Ericaceous shrub abundance (COV_{TS1}, COV_{ES1}, CRH_{TS1}, CRH_{ES1}) and spruce diameter, and a weak positive relationship between first year herb abundance (COV_{H1}) and spruce diameter (Table 4). Using the non-linear regression model (Eq. (3)), CRH_{TS1} had the strongest relationship with spruce diameter, explaining 32% of the variation in third year diameter and 33% of the variation in 10 year diameter (Fig. 3). CRH_{ES1} was a better predictor of spruce diameter than CRH_{TS1} 5 years after manual cutting (R^2 of 0.29 versus 0.21). First year herb cover (COV_{H1}) was positively, but weakly, related to spruce diameter in all years, explaining 4–5% of the variation in 3 and 5 year spruce stem diameter, and 11% in 10 year diameter.

3.7. Competition thresholds

Competition thresholds for year 3 spruce diameter averaged 139 for CRH_{TS1} (range 97–206) and 64% for COV_{TS1} (range 52–90%). An average 25% control spruce and 49% brushed spruce were growing below the CRH_{TS1} thresholds identified for their specific sites. An average 67% of control spruce and 86% of brushed spruce were growing below the COV_{TS1}

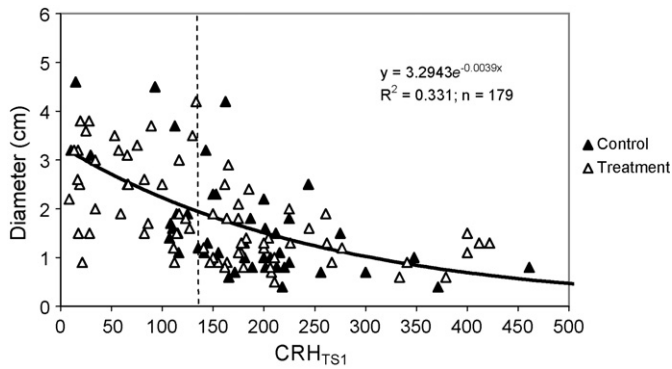


Fig. 3. Fitted non-linear regression model describing the relationship between 10 year spruce diameter and first year CRH_{TS1} . The dashed vertical line represents the average competition threshold for the four sites. Closed and open symbols represent trees in control and treated plots, respectively.

thresholds. Of control spruce that were growing in neighbourhoods above the CRH_{TS1} or COV_{TS1} thresholds, 42 and 48%, respectively, died by year 10. Among brushed spruce growing in neighbourhoods above the CRH_{TS1} or COV_{TS1} thresholds, 28 and 17%, respectively, died by year 10. An average 2% (range 0–5%) of control spruce and 6% (range 0–15%) of brushed spruce achieved at least 80% of the maximum 3-year spruce diameter for their specific site.

4. Discussion

4.1. Spruce responses

In our study, differences in brushed and control spruce survival and growth were evident up to 5 years after manual cutting, but were non-significant by year 10 even though the magnitude of the differences remained relatively constant. The extent of conifer seedling response to vegetation management depends on several factors, including seedling vigour at the time of treatment, the extent to which vegetation competition is affecting conifer growth, and treatment efficacy (Weldon and Slauson, 1986; Wagner and Radosevich, 1998; Cole et al., 2003). In wet, cold climatic environments such as the ESSF zone, treatments that successfully reduce vegetation abundance may not result in large growth improvements due to the presence of other, even stronger, limitations. Conifer seedlings grow slowly at high-elevations (Farnden, 1994) so that differences in size, even where statistically significant, may be small in magnitude. In our study, which was specifically designed to evaluate conifer and vegetation responses under operational conditions, the small differences that occurred in treatment timing and methodology contributed to the variation among replicate sites, thus reducing our power to detect statistically significant differences between treatments.

According to Wagner (2000), thresholds for growth are lower than for survival, and consequently, the effects of competition on survival generally take longer to manifest (Wagner et al., 1989; Simard et al., 2005). This is particularly true of moderately shade-tolerant species such as spruce (Chen, 1997). This did not appear to be the case in our study, where

differences in spruce survival between manually cut and untreated areas were evident as early as the third year post-treatment (96% versus 81%). Differences in survival became non-significant by year 10 despite 18% higher survival in the manual treatment, which is attributed to variability between and within sites. No trends related to basic site factors such as aspect or elevation were apparent, so the variability possibly results from the patchy nature of the Ericaceous shrub community, which has been documented by Caza and Kimmins (1987). Similar differences in survival were observed by Coates et al. (1991) for 3-year-old Engelmann spruce growing amongst an untreated Ericaceous shrub community (76% survival) versus those that were growing where site preparation had reduced vegetation abundance (95% survival). It is uncertain whether or not poor vigour spruce in our study will eventually recover—if they die, survival could decrease by a further 11 and 13% in the control and treatment, respectively.

Other studies reporting long-term spruce survival in non-Ericaceous shrub-dominated communities have focused on lower elevation ecosystems with more moderate climatic conditions. For example, 10 years spruce survival in the temperate Interior Cedar-Hemlock (ICH) zone of southern interior British Columbia was improved by a single manual cutting treatment of a low shrub community (Biring et al., 2003). In contrast, Heineman et al. (2005) compared the effects of brushing herb- and low-shrub communities between medium- and high-elevation sites, and found that conifer responses, including survival, were more strongly affected by climatic factors than by the application of a brushing treatment.

In our study, spruce diameter increased with manual cutting for 5 years post-treatment, but differences were no longer significant by year 10. We attribute the loss of significance to mortality of small, suppressed stems in the control. Biring et al. (2003) report a similar loss of size differences between treated and control Engelmann spruce 7 years after treatment of a lower elevation mixed-shrub community. In contrast, both site preparation and release treatments that reduced abundance of a well-established vegetation community in Alaska improved white spruce (*P. glauca* (Moench) Voss) growth over a 10–11 year period (Cole et al., 2003). In that study, however, survival also declined with brushing because seedlings were exposed to an early freeze.

In our high-elevation study, spruce seedling growth was in the range reported by Lajzerowicz et al. (2006) for 6 year-old Engelmann spruce planted on ESSF sites, but was slower than rates reported elsewhere in British Columbia (Coates et al., 1994). As Farnden (1994) notes, low soil and air temperatures strongly influence seedling survival and growth in the ESSF zone, and naturally regenerated Engelmann spruce may take as long as 20 years to reach 1 m height (Butt and Vyse, 1992). When spruce in our study were 10–12 years old, they averaged 96 cm tall and were growing at approximately 7.5 cm/year. Brushing treatments were applied when spruce were newly planted to 2 years old, which is within the critical period identified by Robinson and Wagner (2006) for brushing black spruce. Those authors suggest that spruce benefit from vegetation removal for up to 5 years after planting. According

to our relative height results, treated spruce were higher in the vegetation canopy than control spruce for the entire 10 year assessment period. Based on work in other vegetation communities (e.g., Comeau, 2002), this suggests that those in the treatment were receiving more light than those in the control.

4.2. Vegetation responses

Ericaceous shrubs are known to recover slowly from cutting (Coates, 1987), and in our study, spot manual cutting reduced shrub cover for at least 10 years, although the decrease was small. Removing the Ericaceous shrub canopy has the potential to increase herb cover on sites where rhizomatous herb species are common (Boateng and Comeau, 1997), but this did not happen on our sites, despite the presence of a variety of rhizomatous species. Herb cover and height were unaffected by the cutting treatment and remained at constant levels throughout the 10 year assessment period, possibly because shrub cover reductions were relatively small.

Manual cutting did not affect species richness in our study, but it caused an increase in species diversity that was evident only in the first year post-treatment. This short-term effect is attributed to a brief decrease in the cover of target shrub species. Simard et al. (2001) also found manual cutting had no effect on richness and diversity of vascular plants in six other vegetation communities in southern interior British Columbia. Haeussler et al. (2002) found little evidence of diversity loss as a result of the conventional range of silvicultural treatments applied to mesic sites in British Columbia—a conclusion that is supported by others (e.g., Sullivan et al., 1998; Boateng et al., 2000; Heineman et al., 2005). However, Haeussler et al. (2002) also reported important shifts in species composition as a result of treatments that removed the forest floor, but not as a result of chemical treatments that left the forest floor intact. Manual cutting, which does not create mineral seedbeds by disturbing forest floor materials, is also likely to have little effect on species composition. Our analysis of species composition indicated that five herb species tended to decrease in treated relative to control plots during the first 5 years of our study, and two other herb species tended to increase. All seven herb species (increasers and decreasers) were rhizomatous, further indicating that spot cutting treatments were not severe enough to stimulate a shift from a shrub- to a rhizomatous herb-dominated community. Three shrub species decreased in relative abundance during the first 5 years after cutting, which was anticipated because they were treatment targets, and three other shrub species increased. All of the shrub species that increased (black gooseberry, elderberry and black huckleberry) were berry-producers that have importance to wildlife (Haeussler et al., 1990).

4.3. Predicting spruce growth from vegetation abundance

We used regression analysis to investigate the predictability of 3, 5 and 10 year spruce stem diameter from first year post-treatment vegetation abundance within a 1.78-m radius. Diameter was chosen because it responds more quickly and

dramatically than height to reductions in vegetation competition (e.g., Lanner, 1985; Lanini and Radosevich, 1986; Simard et al., 2001). The competition index we examined (CRH) used variables that were easily measured in the field. First year shrub cover (COV_{TS1}) and CRH (CRH_{TS1}) were more strongly (negatively) correlated with spruce diameter than were variables based on the abundance of Ericaceous shrubs only, suggesting that less abundant, and usually shorter species were also competing with spruce.

The non-linear model using CRH_{TS1} explained up to 33% of the variation in spruce diameter, with the strongest relationship 10 years after cutting. The relationship was weaker for year 5 diameter (R^2 of 0.206), possibly reflecting the influence of factors other than vegetation competition on spruce vigour. The increase in the moderate R^2 values between years 5 and 10 is consistent with gradual mortality of poor vigour seedlings that were responding to environmental conditions other than vegetation competition. Wagner and Radosevich (1998) found that vegetation abundance explained more variation in Douglas-fir size than we found for spruce, and that R^2 values increased between years 1 and 5 after treatment. In their study, height of the untreated shrub canopy increased during that period, whereas in our study, the untreated Ericaceous shrub community had attained its maximum height before treatment. We expect that the negative correlation between vegetation abundance and spruce diameter will weaken on our sites as surviving spruce outgrow the shrub canopy.

In a lower elevation mixed-shrub community, Comeau et al. (1993) examined the relationship between spruce size and vegetation abundance using the same competition index that we used. That study, which reported unadjusted R^2 values of approximately 98% for the relationship between the competition index and spruce biomass, also identified a strong relationship between the competition index and light transmittance through the vegetation canopy. The lower R^2 values for models predicting spruce diameter from CRH_{TS1} in our study are consistent with our understanding that light competition is only one of several limiting factors affecting spruce growth. Manual cutting reduces vegetation cover and height, but it does not ameliorate the effects of cold soil conditions and short growing seasons.

Weldon and Slauson (1986) define competition in terms of both importance (as defined by the adjusted R^2 value), and intensity (as defined by the slope of the linear regression model). On our study sites, the slope and R^2 values for a linear model (data not shown) suggested that competition was of low intensity and low importance 3 years post-treatment. By year 10, they indicated that intensity and importance had both increased slightly, suggesting that vegetation competition was still affecting surviving seedlings growing in dense patches of vegetation (probably the 10–13% of seedlings that were of poor vigour).

4.4. Competition thresholds

Competition thresholds for year 3 spruce stem diameter were high and indistinct on our sites, and individual values

varied considerably around the means of 139 for CRH_{TS1} and 64% for COV_{TS1} . On all sites, spruce diameter varied widely at low and medium values of CRH_{TS1} and COV_{TS1} , supporting our hypothesis that vegetation competition was only one of several constraints to growth (e.g., Burton, 1993) on these high-elevation sites where air and soil temperature limitations are severe (Farnden, 1994). Of spruce that were growing above the CRH_{TS1} and COV_{TS1} thresholds, approximately one-third died by year 10. In another study that compared competition thresholds in other (non-Ericaceous) shrub- and herb-dominated communities, Heineman et al. (2005) found that competition thresholds were lower on high-elevation ESSF zone sites than on more temperate medium-elevation ICH zone sites, possibly because seedlings on the high-elevation sites were less vigorous and had lower tolerance to competition.

5. Conclusions and management implications

Ten years after spot manual cutting treatments were applied to release Engelmann spruce seedlings from the Ericaceous shrub community on high-elevation sites in southern interior British Columbia, we cannot conclude that the treatment improved either conifer survival or growth. Despite 18% lower survival in the untreated control than the treatment, combined with a greater proportion of poor vigour trees destined to die, responses were sufficiently variable among sites that survival differences were not statistically significant. The Ericaceous shrub community is naturally patchy (Caza and Kimmins, 1987), which may have contributed to the variable results, and our regression results also indicate that vegetation competition was not the primary constraint on spruce performance. Shrub abundance accounted for only 33% of the variation in spruce diameter, suggesting that other factors, perhaps related to the wet, cold environment, were more limiting. Similar conclusions were drawn in a related study that compared brushing effects between high- and medium-elevation sites (Heineman et al., 2005).

Our variable results suggest that the cost of applying spot manual cutting treatments to facilitate Engelmann spruce establishment may not be warranted. Current operational practices that include harvesting in small openings, mechanical or chemical site preparation, early and repeated brushing, and the selection of large stock types can contribute to successful seedling establishment on high-elevation sites (Wood and von Althen, 1993; Thiffault et al., 2003), and reduce the likelihood that brushing will be required (Boateng and Comeau, 1997). However, circumstances still exist (e.g., backlog areas created by failed treatments or natural disturbance) where management of the fully developed Ericaceous shrub community is necessary. The magnitude of the difference in survival between the control and treatment in our study cannot be completely ignored, despite the lack of statistical significance. Further work to identify the characteristics of sites where brushing treatments improve survival is warranted.

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